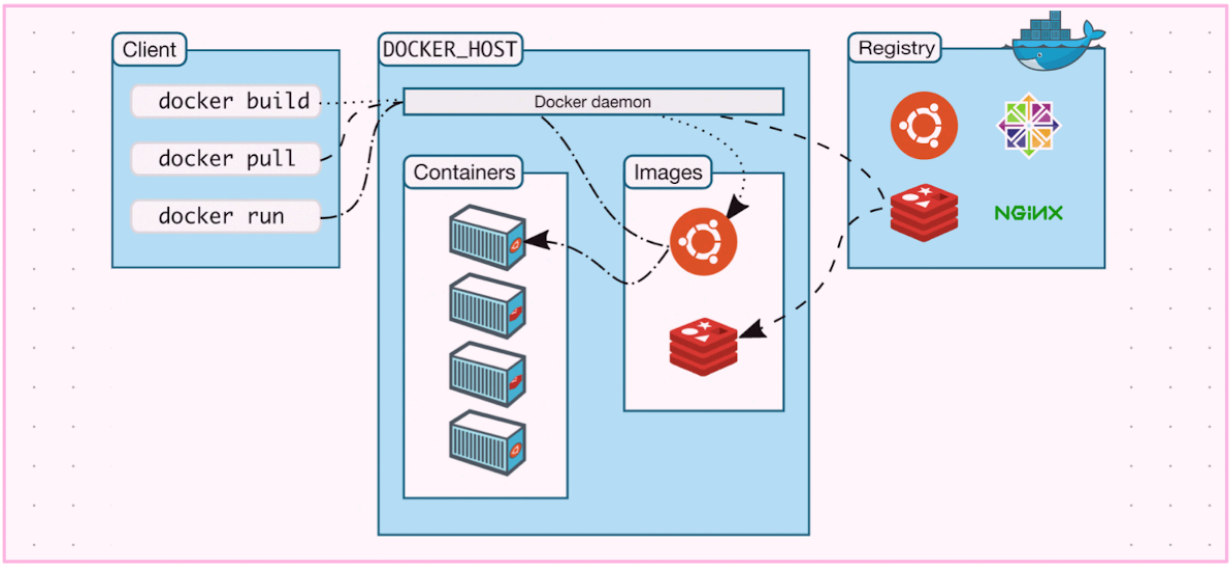
Docker Architecture

The Docker architecture employs a client-server design. A Docker client talks to the Docker daemon via the REST API and the daemon manages the containers according to the commands issued by the user. The REST API is in charge of communication between the client and the daemon.

Let's take a deeper look at Docker's architecture; how it runs applications in isolated containers and how it works under the hood.



There are five major components in the Docker architecture:

a) Docker Daemon listens to Docker API requests and manages Docker objects such as images, containers, networks and volumes.

b) Docker Clients: With the help of Docker Clients, users can interact with Docker. Docker client provides a command-line interface (CLI) that allows users to run, and stop application commands to a Docker daemon.

c) Docker Host provides a complete environment to execute and run applications. It comprises of the Docker daemon, Images, Containers, Networks, and Storage.

d) Docker Registry stores Docker images. Docker Hub is a public registry that anyone can use, and Docker is configured to use images on Docker Hub by default. You can run your own registry on it.

e) Docker Images are read-only templates that you build from a set of instructions written in Dockerfile. Images define both what you want your packaged application and its dependencies to look like what processes to run when it's launched.